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SUPERSEDING
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MILITARY SPECIFICATION

CABLE ASSEMBLY, SPECIAL PURPOSE, ELECTRICAL CX-11230( )/G AND CABLE ASSEMBLY ADAPTER CX-10734( )/G

1. SCOPE

1.1 Scope.- This specification covers Cable Assembly, Special Purpose, Electrical CX-11230( )/G using Cable WD-37( )/U, having a twisted pair of coaxial cables with overall braid and jacket, terminated at each end with a coaxial hermaphroditic watertight connector UG-1870( )/U. The cable assembly furnishes the circuit for the 12, 24, and 48 Channel Pulse Code Modulation Multiplex equipment. It also covers Cable Assembly Adapter CX-10734( )/G using cable WD-37( )/U with connector UG-1870( )/U on one end and one male connector UG-1872( )/U and one female connector UG-1871( )/U on second end.

2. APPLICABLE DOCUMENTS

2.1 Documents.- The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-561	Solder, Silver.

MILITARY

MIL-M-13231	Marking of Electron Items.
MIL-F-14072	Finish for Ground Signal Equipment.
MIL-C-55442	Cable Assemblies and Cord Assemblies, Packaging and Packing of.

FSC 5995

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-810	Environmental Test Methods.

DRAWINGS

ELECTRONICS COMMAND

SC-GL-323276	Connector Plug, Electrical UG-1872( )/U.
SC-GL-323277	Connector Plug, Electrical UG-1871( )/U.
SC-GL-323378	Connector Plug, Electrical UG-1870( )/U.
DL-SC-A-339551	Cable Assembly Adapter CX-10734( )/G
SM-D-339552	Cable Assembly Adapter CX-10734( )/G.
SM-C-434025	Cable Assembly, Special Purpose, Electrical CX-11230( )/G.
SM-B-434064	Cable Assembly Instructions.
SM-A-434065	Cable, Special Purpose, Electrical WD-37( )/U.
SC-DL-434067	Cable Assembly, Special Purpose, Electrical CX-11230( )/G.
SC-D-551562	Connector Assembly Instructions (Female).
SC-D-551575	Connector Assembly Instructions (Male).

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Construction.- Cable Assembly, Special Purpose, Electrical CX-11230( )/G shall be constructed in accordance with the requirements of this specification, and of SC-DL-434067. Cable Assembly Adapter CX-10734( )/G shall be constructed in accordance with the requirements of this specification, and of DL-SC-A-339551.

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3.1.1 Cable assembly instructions.- Assembly of the connector to the cable shall be in accordance with SM-B-434064 for Cable Assembly CX-11230( )/G. Assembly of the connectors to the cable shall be in accordance with SM-B-434064, SM-D-551562 and SM-D-551575 for Cable Assembly CX-10734( )/G.

3.2 Materials.- Only virgin material shall be used in the manufacture of insulation or jacket compounds used in the fabrication of cable and wire.

3.3 Soldering.- Soldering used for electrical connections shall conform to Requirement 5 of MIL-STD-454.

3.4 Cleaning.-

3.4.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled.

3.4.2 Units.- After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.5 Finish.- Connectors shall be given protective finish in accordance with MIL-F-14072 and equipment drawings. (See 4.4).

3.6 Marking.- Marking shall conform to MIL-M-13231. (See 4.4).

3.7 Interchangeability.- Like units, and assemblies shall be physically and functionally interchangeable, without modification of such items. Individual items shall not be handpicked for fit or performance. Where a listed dimension is not within design limits, it shall be considered a major defect. (See 4.10).

3.8 Individual coaxial cable.-

3.8.1 Dimensional stability.- The average protrusion of the inner conductor from its dielectric core shall not exceed 1/32 inch. (See 4.6.1).

3.8.2 Flow.- The inner conductor shall not be displaced from the center of the cable by more than 15 percent of the diameter of the dielectric. Measure along the direction of flow when tested in accordance with 4.6.2.

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3.8.3 Tubing.- Neither the dielectric nor the jacket shall show evidence of cracking when tested in accordance with 4.6.3. After examination of the jacket, the jacket and braid shall be removed and the dielectric examined.

3.8.4 -40°C bend.- Neither the dielectric nor the jacket shall show evidence of cracking after being tested in accordance with 4.6.4. After examination of the jacket, the jacket and braid shall be removed and the dielectric examined.

3.8.5 -55°C bend.- Neither the dielectric nor the jacket shall show evidence of cracking after being tested in accordance with 4.6.5. After examination of the jacket, the jacket and braid shall be removed and the dielectric shall be examined.

3.8.6 Low temperature after aging.- The individual coaxial cable shall meet the requirements of 3.8.5 after being tested in accordance with 4.6.6.

3.8.7 Impedance.- The characteristic impedance of the individual coaxial cable shall be a maximum of 65 ohms and a minimum of 55 ohms in the frequency range from 400kc to 1000kc. In the frequency range from 1000kc to 3000kc, the characteristic impedance of the cable shall be a maximum of 60 ohms and a minimum of 55 ohms. The phase angle shall be 4° maximum in the 1000-3000kc range only. (See 4.6.7).

3.9 Connectors.-

3.9.1 Rotation.- The torque required to rotate the coupler of the UG-1870( )/G shall not exceed 0.75 inch pounds and the torque required to rotate the boot subassembly of the UG-1872( )/U shall not exceed 2.0 inch pounds when tested in accordance with paragraph 4.7.1.

3.9.2 Salt spray.- After subjection to the test specified in 4.7.2, the plating shall remain intact and the connectors shall show no evidence of deterioration and meet the requirements of 3.9.1.

3.9.3 Shock drop.- After being subjected to the test specified in 4.7.3, there shall be no damage and/or loosening of parts and the connectors shall be mechanically operable. Any physical damage shall be minor. Continuity shall be continuous and the electrical requirements of 3.10.1 Dielectric withstanding voltage and 3.10.2 Insulation resistance shall be met.

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3.9.4 Shock.- The coupled cable assembly shall be capable of withstanding the shock test specified in 4.7.4 without loosening of the coupling, damage and/or loosening of parts and the connectors shall be mechanically operable. Continuity shall be continuous and the electrical requirements of 3.10.1 Dielectric withstanding voltage and 3.10.2 Insulation resistance shall be met.

3.10 Cable Assemblies CX-11230( )/G and CX-10734( )/G.-

3.10.1 Dielectric withstanding voltage.- Each coaxial cable of the cable assemblies shall withstand a potential of 2,500 volts dc for one minute between the center contact and the outer contact when tested in accordance with 4.8.1.

3.10.2 Insulation resistance.- Each coaxial cable of the cable assemblies shall have an insulation resistance of not less than 50,000 megohms between the center contact and the outer contact when tested in accordance with 4.8.2.

3.10.3 DC resistance.- The direct current resistance of the center conductor and contacts of each coaxial cable shall not exceed 22 ohms per 1,320 feet at or corrected to 20°C. The direct current resistance of the braid and outer contact of each coaxial cable shall not exceed 7.5 ohms per 1,320 feet at or corrected to 20°C when tested in accordance with 4.8.3. The 100 foot coaxial cables direct current resistance shall be prorated from the above with a 10% tolerance on the calculated resistance. The Cable Assembly Adapter CX-10734( )/G shall be checked for electrical continuity by a point-to-point method. The value shall be not greater than 0.1 ohms for the center conductor and contacts and .07 ohms for the braid and outer contact.

3.10.4 Attenuation.- The attenuation shall not exceed 8.5 db per mile at 100kc and shall not exceed 38 db per mile at 2,304kc when tested in accordance with 4.8.4.

3.10.5 Crosstalk.- The near end crosstalk level on the center conductor of one coaxial element shall be at least 67 db below the signal level impressed on the center conductor of the other coaxial element. (See 4.8.5).

3.10.6 Water immersion.- There shall be no deterioration or evidence of water leakage in the connector assembly and the insulation resistance shall be not less than 50,000 megohms when tested in accordance with 4.8.6.

3.10.7 Moisture resistance.- After being subjected to the test specified in 4.8.7, the assemblies shall be electrically continuous and meet the requirements of 3.10.1 Dielectric withstanding voltage and 3.10.2 Insulation resistance except that the IR shall be a minimum of 100 megohms during cycling and a minimum of 10,000 megohms after 24 hours drying. The connectors shall show no evidence of deterioration.

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3.10.8 Temperature cycling.- During the test specified in 4.8.8, the equipment shall be electrically continuous and meet the requirements of 3.10.1 Dielectric withstanding voltage, 3.10.2 Insulation resistance, 3.10.3 DC resistance, and shall show no evidence of deterioration.

3.10.9 Vibration.- After being subjected to the test specified in 4.8.9, there shall be no loosening of the coupling, damage and/or loosening of parts. Continuity shall be continuous and the electrical requirements of 3.10.1 Dielectric withstanding voltage and 3.10.2 Insulation resistance shall be met.

3.10.10 Cable retention withstanding.- While being subjected to the test specified in 4.8.10, there shall be no loss of continuity of either center conductor or conductor braid and the cable and the assemblies shall meet the requirements of 3.10.1 Dielectric withstanding voltage, 3.10.2 Insulation resistance, and the assembly shall be electrically continuous. The test shall continue for a sufficient length of time to perform the electrical tests. The locking devices shall not become detached from the connectors.

3.11 Cable.-

3.11.1 Breaking strength.-

3.11.1.1 Static.- The finished cable shall be visually examined for damage and meet the requirement of 3.10.3 when tested in accordance with 4.9.1.1.

3.11.1.2 Dynamic.- The finished cable shall be visually examined for damage and meet the requirement of 3.10.3 when tested in accordance with 4.9.1.2.

3.11.2 Spark test.- The dielectric and the jackets shall withstand the spark test using the voltages indicated in paragraph 4.9.2.

3.12 Repairs.- Repairs shall conform to the following requirements.

3.12.1 Conductor strand splices.- When splices are required in the individual copper strands, they shall be the butt-brazed type using silver solder in accordance with QQ-S-561. If the brazed splice is larger than the original wire, it shall be carefully dressed down to proper size which shall be not less than that of the original wire. Seven strand splices may be made in the stranding operation by joining the individual strands as described above provided that the individual strand splices are spaced at least 1 foot apart. No strands shall be kinked, and the smoothness of lay shall not be altered. When a splice is made in this manner, it shall be possible to move a die having a diameter of 0.030 inch and a length of 1 inch along the length of the spliced conductor without binding or necessity for the exertion of a definite pressure.

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3.12.2 Insulation repairs.- Insulation repairs shall be made with a suitable heat stabilized, foamed high density polypropylene compound and shall be of the heat molded type. In making such repairs, the original insulation shall be removed by smooth tapering cuts, extending on each side of the defect without nicking the conductor. The finished repair shall be cylindrical and smooth with no flash of excess patching compound. The overall diameter and minimum wall thickness as specified on SM-A-434065 shall be maintained as closely as possible. The maximum length of a repair shall be 4 inches. Insulation repairs shall be sufficiently flexible to meet the low temperature requirements of 3.8.4 and 3.8.5 using a mandrel of 0.225 ± 0.005 inch. There shall be no evidence of cracking of the patch or of the joint between the patch and the original insulation.

3.12.3 Jacket repairs.- If repairs are required in the jackets, the materials and methods used shall be subject to the approval of the contracting officer. Only 2 repairs are permitted in a 1/4 mile length. Not more than 15% of the total 1/4 mile lengths on order may contain repairs. Not more than 15% of any individual shipment may contain repairs. Cable assemblies less than 1/4 mile in length shall contain no jacket repairs.

3.12.4 Cable splice.- No cable splices are permitted in Cable Assembly, Special Purpose, Electrical CX-11230( )/G or Cable Assembly Adapter CX-10734( )/G.

3.13 Workmanship.- Cable Assembly, Special Purpose, Electrical CX-11230( )/G and Cable Assembly Adapter CX-10734( )/G shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein: (See 4.11).

3.3	Soldering
3.4	Cleaning
3.12	Repairs

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections.- The inspection requirements specified herein are classified as follows:

(a) First article inspection. (See 4.3). Does not include preparation for delivery.

(b) Inspections covered by subsidiary documents. (See 4.4).

(c) Quality conformance inspections.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5).

(2) Quality conformance inspection of preparation for delivery. (See 4.12).

4.3 First article.- Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units.- The contractor shall furnish the number of first article units in accordance with the following:

(a) Cable Assembly, Special Purpose, Electrical CX-11230( )/G:

(1) Six (6) each CX-11230( )/G.

(2) Six (6) each UG-1870( )/U.

(3) Fifty (50) feet WD-37( )/U.

(4) One hundred (100) feet individual coaxial cable.

(b) Cable Assembly Adapter CX-10734( )/G:

(1) Twelve (12) each CX-10734( )/G.

(2) Six (6) each UG-1870( )/U.

(3) Six (6) each UG-1871( )/U.

(4) Six (6) each UG-1872( )/U.

(5) Fifty (50) feet WD-37( )/U.

(6) One hundred (100) feet individual coaxial cable.

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4.3.2 First article inspection.- The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4 and the inspections specified in Table I.

4.3.3 First article data.- The first article test plan and test report(s) shall be as required in the contract or purchase order.

Table I.- First article inspection

Inspection	Req. Para.	Test Para.	
1. Inspection covered by subsidiary documents		4.4	Inspection to be performed on all units
2. Group A inspection	See Table II		Inspection to be performed on all units
3. Group B inspection	See Table III		Inspection to be performed on all units
4. Group C inspection			
Subgroup 1	See Table IV		Inspection to be performed as specified in test paragraphs
Subgroup 2	See Table		Inspection to be performed as specified in test paragraphs
Subgroup 3	See Table		Inspection to be performed as specified in test paragraphs

4.4 Inspections covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where required
Finish	3.5
Marking	3.6

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4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspections specified in 4.4 and 4.5.1 through 4.5.3. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection.- Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQLs indicated in Table VII.

4.5.1.1 Order of inspection within Group A.- Group A inspection shall be performed in an order satisfactory to the Government.

Table II.- Group A inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and mechanical	3.13	4.11	1.0%	4.0%
Dielectric withstanding voltage	3.10.1	4.8.1)		
Insulation resistance	3.10.2	4.8.2)	1.0%	*
DC resistance	3.10.3	4.8.3)	combined	
Attenuation	3.10.4	4.8.4)		
Spark test	3.11.2	4.9.2	1.0%	*

* All electrical and spark test defects are considered major.

4.5.2 Group B inspection.- Group B inspection shall normally be performed in inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans.- The Group B AQLs shall be 1.0 percent defective and the inspection levels shall be S-3.

4.5.2.2 Order of inspection within Group B.- Group B inspection shall be in an order satisfactory to the Government.

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Table III.- Group B inspection

Inspection	Req. Para.	Insp. Para.
Interchangeability	3.7	4.10
Rotation	3.9.1	4.7.1
Water immersion	3.10.6	4.8.6
Cable retention withstanding	3.10.10	4.8.10

4.5.3 Group C inspection.- Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table IV. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspection.-

4.5.3.1.1 Subgroup 1.- For this subgroup, specimens or cable assembly as specified in the test paragraphs shall be selected from the first units produced or first production lot. For subsequent Group C inspection cable specimens or cable assembly shall be selected each week or every 100 miles of finished cable produced, whichever comes first.

4.5.3.1.2 Subgroup 2.- For this subgroup, four (4) connectors shall be selected from the first units produced or first production lot. For subsequent Group C inspection four (4) connectors shall be selected every 3 months or 1,500 units, whichever comes first.

4.5.3.1.3 Subgroup 3.- For this subgroup, four (4) cable assemblies shall be selected from the first production lot. For subsequent Group C inspection four (4) cable assemblies shall be selected every 6 months or every 6,000 units, whichever comes first. Tests shall be conducted in the order listed.

4.5.3.2 Order of inspection within Group C.- Group C inspection shall be performed in an order which is satisfactory to the Government except for Subgroup 3.

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Table IV.- Group C inspection

Inspection	Req. Para.	Insp. Para.
Subgroup 1		
Dimensional stability	3.8.1	4.6.1
Flow	3.8.2	4.6.2
Tubing	3.8.3	4.6.3
-40°C bend	3.8.4	4.6.4
-55°C bend	3.8.5	4.6.5
Low temperature after aging	3.8.6	4.6.6
Impedance	3.8.7	4.6.7
Crosstalk	3.10.5	4.8.5
Breaking strength	3.11.1	4.9.1
Subgroup 2		
Salt spray *	3.9.2	4.7.2
Shock drop	3.9.3	4.7.3
Shock	3.9.4	4.7.4
Vibration	3.10.9	4.8.9
Subgroup 3		
Moisture resistance	3.10.7	4.8.7
Temperature cycling	3.10.8	4.8.8

* The equipment shall be thoroughly washed, cleaned, dried and refurbished after this inspection before proceeding with subsequent inspection.

4.5.3.3 Group C failures.- Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group C sample units.- Unless otherwise specified, sample units which have been subjected to Breaking strength, Moisture resistance and Temperature cycling, and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are re-subjected to and pass Group A inspection only.

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4.5.5 Disposition of Group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or purchase order, provided sample units meet requirements of 4.5.4.

4.6 Individual coaxial cable.-

4.6.1 Dimensional stability.- The dimensional stability of 3 - 10 foot sections of the individual coaxial cable shall be determined as follows:

(a) Make a square, clean cut through the cable at each end of the 10 foot sections.

(b) Place the cable sections, coiled or straight, in an oven for 20 hours at 85°C ± 2°.

(c) Remove the cable sections from the oven and cool to room temperature.

(d) Measure both ends of the cable sections for inner-conductor protrusion.

(e) The results of the measurements shall be averaged on each section and shall conform to 3.8.1.

4.6.2 Flow.- Three specimens of the individual coaxial cable, each 30 times the diameter of the cable in length, shall be prepared for testing by skinning at least 1 inch at each end to the bare conductor(s). Each end of the inner conductor of a coaxial cable shall have a 6 lb. weight applied. The specimens shall be draped symmetrically, with the weights hanging freely, over a horizontal test mandrel whose diameter is 10 times the outside diameter of the cable, and placed in an oven maintained at 98°C ± 2° for 7-1/2 hours. The weights then shall be removed and the cable, preferably supported by the mandrel, shall be removed from the oven and allowed to cool to room temperature. The cable shall not be straightened. Solid dielectric cable shall either be cut at the middle of the bend with a jeweler's saw, or an x-ray photograph shall be taken. Measurements shall be taken to determine conformance to 3.8.2.

4.6.3 Tubing.- Three specimens of the individual coaxial cable shall be looped back and then wound tightly on itself for 5 close turns, the ends securely taped, and the specimens placed in an oven maintained at a temperature of 95°C ± 2° for 1 hour. After removal from the oven, the specimens shall be examined under a magnifying glass of approximately 3 diameters magnification (focal distance 3 inches) and meet the requirements of 3.8.3.

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4.6.4 -40°C bend.- Three specimens of the individual coaxial cable and a test mandrel whose diameter shall be 0.500 inch ± 0.005 inch shall be placed in a cold chamber maintained at a temperature of -40°C ± 1° for 24 hours. At the end of this time, and while still in the cold chamber, the cable shall be wound around the mandrel for 7 close turns at a rate of approximately 1 turn per second and meet the requirements of 3.8.4 when examined under a magnifying glass of approximately 3 diameters magnification (focal distance 3 inches). In conducting this test, the first and last turns shall be discounted and no object having a higher temperature than -40°C ± 1° shall come within 12 inches of the part of the specimen which is under test.

4.6.5 -55°C bend.- Three specimens of the individual coaxial cable and a test mandrel whose diameter shall be 10 x the diameter of the cable shall be placed in a cold chamber maintained at a temperature of -55°C ± 1° for 24 hours. At the end of this time, and while still in the cold chamber, the cable shall be wound around the mandrel for 3 close turns at 15 rpm and meet the requirements of 3.8.5 when examined under a magnifying glass of approximately 3 diameters magnification (focal distance 3 inches). In conducting this test, the first and last turns shall be discounted and no object having a higher temperature than -55°C ± 1° shall come within 12 inches of the part of this specimen which is under test.

4.6.6 Low temperature after aging.- Six specimens of individual coaxial cable shall be aged for 7 days at + 80°C ± 1°, followed by a cooling period of at least 24 hours at room temperature. The cable specimens shall then be submitted to and meet the requirements of 3.8.6.

4.6.7 Impedance.- To determine conformance with the requirements of 3.8.7, the characteristic impedance shall be measured on a 25 foot length of individual coaxial cable in the range of 400 to 3000kc. Measurements are required at 400kc, 1000kc, 2300kc, and 3000kc. Measurements at other points within the 400kc to 3000kc range are subject to spot checking.

4.7 Connectors.-

4.7.1 Rotation.- A torque shall be applied to the coupler of the UG-1870( )/U connectors and the boot subassembly of the UG-1872( )/U connectors about the axis of symmetry. The torque shall be measured with a torque wrench to determine conformance to 3.9.1.

4.7.2 Salt spray.- Two unmated connectors not assembled to cable shall be subjected to the salt spray (corrosion) test in accordance with MIL-STD-202, Method 101, Condition B, and meet the requirements of 3.9.2.

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4.7.3 Shock drop.- Two unmated connectors UG-1870( )/U, each wired to 5 feet of cable, shall be dropped at random six times with protective caps coupled to the connectors, and six times with protective caps dangling from a height of ten feet onto 2 inch fir wood backed by concrete to determine conformance to the requirements of 3.9.3. The connectors shall be visually examined and the caps tightened after each drop. The CX-10734( )/G shall be dropped in the same manner and meet the same requirements.

4.7.4 Shock.- Two coupled connectors UG-1870( )/U, each wired to 5 feet of cable and the CX-10734( )/U shall be tested in accordance with MIL-STD-202, Method 205, Test Condition C, or Method 202 at equivalent acceleration to determine compliance with the requirements of 3.9.4.

4.8 Cable Assemblies CX-11230( )/G and CX-10734( )/G.-

4.8.1 Dielectric withstanding voltage.- The cable assemblies shall be tested in accordance with Method 301 of MIL-STD-202 to determine conformance to 3.10.1.

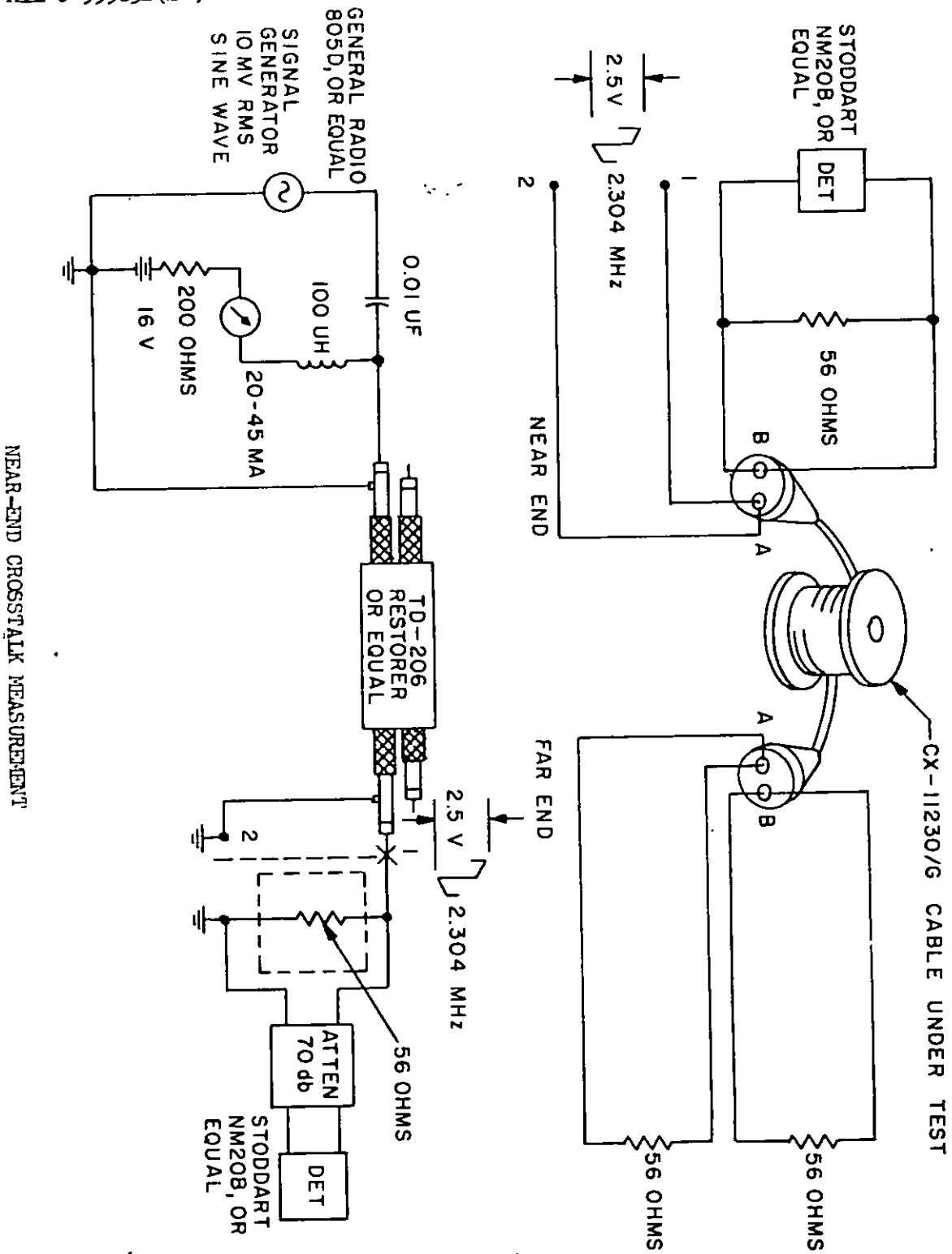
4.8.2 Insulation resistance.- The cable assemblies shall be tested in accordance with Method 302, Condition B, of MIL-STD-202 to determine conformance to 3.10.2.

4.8.3 DC resistance.- The cable assemblies shall be tested in accordance with Method 303 of MIL-STD-202 to determine conformance to 3.10.3.

4.8.4 Attenuation.- The attenuation shall be measured on 4 one-quarter mile cable assemblies connected in series, to determine conformance to 3.10.4. Measurements shall also be made in the range of 10kc to 3000kc to establish the attenuation characteristics.

4.8.5 Crosstalk.- The near end crosstalk shall be measured on completed assemblies wound on the reel, using a pulse pattern operation at a 2.304mc rate with a dipulse waveform at a signal amplitude of 2.5 volts peak-to-peak, and with the far end of the coaxial elements terminated in their characteristic impedance. The nominal width of the pulse at the peak amplitude shall be 217 nanoseconds. See Figure 1 for test set-up. Restorer shall be Government-furnished. (See 3.10.5).

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NEAR-END CROSSTALK MEASUREMENT

Figure 1

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4.8.6 Water immersion.- The connectors on the cable assemblies shall be immersed to a depth of 3 feet of water for 16 hours to determine compliance with the requirements of 3.10.6 Fifty percent of the connectors shall be mated, twenty-five percent shall be with protective cover in place, and twenty-five percent shall be without protective cover. When the sample size is not a multiple of four (4), the odd connector(s) shall be tested with protective cover(s) in place.

4.8.7 Moisture resistance.- Two cable assemblies shall be tested in accordance with Method 507, Procedure II, of MIL-STD-810. Prior to testing, the cable assemblies shall meet full specification performance for the measurements specified in 3.10.7. Measurements shall be made during the periods shown on the figure for the cycle. Do not move cable assemblies from the humidity chamber for measurements. No repair or replacement of parts shall be permitted. If the cable assemblies fail to meet the specified requirements during cycling or after final conditioning, they do not pass the test.

4.8.8 Temperature cycling.- Two cable assemblies shall meet the requirements of 3.10.8, after subjection to the temperature cycle of Table V.

Table V.- Temperature cycle

Step	Temperature (C)	Time (Hours)
1	85 ± 2	4
2	$25 \pm {10 \atop 5}$	4
3	-55 ± 2	4
4	$25 \pm {10 \atop 5}$	4

4.8.9 Vibration.- Two coupled connectors UG-1870( )/U, each assembled to 5 feet of cable and/or two CX-10734( )/G shall be tested in accordance with MIL-STD-202, Method 204, Test Condition A, and meet the requirements of 3.10.9.

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4.8.10 Cable retention withstanding.- A completed Cable Assembly CX-11230( )/G shall be coupled to a test connector acceptable to the Government. The assembly shall be subjected to an instantaneous tensile force of 550 pounds minimum. The instantaneous force shall be applied by attaching a 200 pound weight to the cable by a means which is not destructive to the cable, and releasing the weight in such a manner as to produce the 550 pound instantaneous load. A calibrated tensiometer or other recording device acceptable to the Government shall be used to measure the instantaneous load. The load shall be recorded. The assembly shall be subjected to the instantaneous load three (3) consecutive times. After the third application of the load, the 200 pound weight shall remain suspended for the time required to determine compliance with the requirements of 3.10.10. The Cable Assembly Adapter CX-10734( )/G shall be subjected to a 45 pound load on each connector UG-1871( )/U and UG-1872( )/U with the U-1870( )/U attached to the test connector described above. The assembly shall meet the requirements of 3.10.10.

4.9 Cable.-

4.9.1 Breaking strength.-

4.9.1.1 Static.- One end of the cable shall be anchored securely. A static tensile force of 550 pounds shall be applied axially for 1 minute and shall meet the requirements of 3.11.1.1.

4.9.1.2 Dynamic.- The cable shall be secured as in 4.9.1.1. The free end of the cable shall be subjected to an instantaneous tensile force of 550 pounds in an axial direction and meet the requirements of 3.11.1.2.

4.9.2 Spark test.- The duration of time that any one point of the cable is subjected to the specified voltage shall be not less than 1/4 second. Any spark test failure shall be repaired and resparked. The finished cable and parts thereof shall meet the requirements of 3.11.2 using the voltages indicated below:

(a) Spark test the dielectric over the conductor at 3.5 kv rms.

(b) Spark test the individual coaxial jacket at 2.3 kv rms.

(c) Spark test the twisted pair of coaxial cables at 2.3 kv rms.

(d) Spark test the outer jacket of the completed cable at 6.3 kv rms.

4.10 Inspection for interchangeability.- The connectors on the cable assemblies shall be gaged using the applicable gages listed on Gage Lists SC-GL-323276, SC-GL-323277 and SC-GL-323378 to determine conformance to the physical interchangeability requirements of 3.8.

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4.10.1 Cable Assembly CX-11230( )/G.- Cable assembly dimensions shall be measured to determine conformance with SM-A-434025.

4.10.2 Cable Assembly CX-10734( )/G.- Cable assembly dimensions shall be measured to determine conformance with SM-D-339552.

4.10.3 Cable.- Cable dimensions, braiding, twisting and shield shall conform to SM-A-434065.

4.11 Visual and mechanical inspection.- Cable Assembly, Special Purpose, Electrical CX-11230( )/G and Cable Assembly Adapter CX-10734( )/G shall be examined for defects listed in Table VI. (See 3.13).

Table VI.- Classification of visual and mechanical defects

Classification	Defects
Major	Threads chipped, broken or stripped. Rubber parts cut or ripped. Coupler not operating freely. Substandard plating. (Flaking, peeling, blisters, etc.) Contacts missing, broken or bent. Any foreign objects or material preventing the mating of connectors. Cut or ruptured cable. Finish not as specified. Insulators cracked or broken.
Minor	Scratches, cuts, abrasions, etc. with exposure of bare metal. Abrasions on rubber parts. Abrasions on cable jacket.

4.12 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be in accordance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Package and pack in accordance with MIL-C-55442(EL).

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6. NOTES

6.1 Intended use.- Cable Assembly, Special Purpose, Electrical CX-11230( )/G is used in communication systems utilizing 12, 24, or 28 PCM channels. Cable Assembly Adapter CX-10734( )/G is used to connect the CX-11230( )/G to the Restorers TD-206 to the Cable Assemblies CX-4245( )/G or to the equipment.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of document.

(b) Type required.

(c) Level of packaging and level of packing required for shipment. (Level A, level B or level C).

(d) The specific paragraphs of MIL-C-55442(EL) which are applicable to the particular procurement.

(e) Place of final inspection.

6.3 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design: for example, CX-11230W/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1).

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3).

CUSTODIAN

ARMY EL

Preparing Activity

ARMY EL

Project No. 5995-A091

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-55[illegible]B(E) CABLE ASSEMBLY, SPECIAL PURPOSE, ELECTRICAL CX-11230( )/G CABLE ASSE[illegible] ADAPTER CX-10730( )/G

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "Yes", in what way?)*

4. *REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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